THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE CHAPTER TEST SET A

Chapter 2: **Electricity: Circuits and Their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A combination of two or more electric cells is called a: 1

(a) circuit **(b)** switch **(c)** filament **(d)** battery

2. A circuit in which the current flows continuously is a: 1

(a) closed circuit **(b)** broken wire **(c)** open circuit **(d)** short key

3. In the symbol of a cell, the longer line represents the: 1

(a) switch **(b)** bulb

(c) positive terminal **(d)** negative terminal

4. Which of the following is an insulator? 1

(a) Graphite **(b)** Aluminium foil **(c)** Plastic **(d)** Iron nail

5. An LED glows only when its: 1

(a) connected to no cell

(b) both terminals reversed

(c) legs are cut

(d) positive terminal is connected to the positive side of the cell

6. The current from a cell is: 1

(a) mains current **(b)** high voltage AC

(c) direct current (DC) **(d)** alternating current

7. Which is a safe source for experiments in school? 1

(a) Cells **(b)** High-voltage line

(c) Mains socket **(d)** Generator

8. Lemon juice can conduct because it: 1

(a) is colourless **(b)** is cold

(c) is sweet **(d)** contains dissolved substances

9. In a circuit diagram, a bulb is shown by a: 1

(a) triangle **(b)** straight line

(c) circle with a cross or loop **(d)** square

10. Which of these makes up a circuit? 1

(a) Only a switch **(b)** Cell, wires, bulb and switch

(c) Only a bulb **(d)** Only a cell

MASTERMIND

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 15–25 words each.

11. What is an electric circuit? 2

12. Differentiate between a cell and a battery. 2

13. Why does a bulb not glow when the switch is open? 2

14. Name two conductors and two insulators. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe how to make a simple electric circuit with a cell, bulb, wire and switch. 3

16. Draw the circuit diagram of a cell, a bulb and a switch (describe) and mark the direction of current. 3

17. How can you test whether a material is a conductor? Describe the activity. 3

OR

Classify: copper, rubber, iron, plastic, aluminium, wood as conductors and insulators. 3

18. Why do electricians use rubber gloves and wooden-handled tools? 3

OR

A bulb does not glow even after all connections are made. List three possible reasons. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) Define a conductor and an insulator. 5
(ii) List two examples of each.
(iii) Describe an activity to test materials.

OR

Case study: Riya finds that her torch will not light. She checks the cell, bulb and switch. 5
(i) Draw the circuit components.
(ii) List three possible faults.
(iii) How can she test the bulb?
(iv) Which is a safe energy source for her?

20. (i) What is a circuit diagram? 5
(ii) Draw the circuit of a cell, two bulbs and a switch.
(iii) Explain how to identify the positive terminal.

OR

Case study: An electrician repairs a fan with rubber gloves and a plastic-handled screwdriver. 5
(i) Why?
(ii) Which materials are conductors?
(iii) What if the floor is wet?
(iv) Suggest safety steps.

END OF QUESTION PAPER — Total: 40 Marks